Working with AI

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This site collects the UCD-SERG lab's notes on working responsibly and effectively with AI coding assistants: what they are, how to use them, and the policies lab members follow when using them. It was migrated out of the UCD-SERG Lab Manual's "Working with AI" chapter, which had grown large enough to deserve a dedicated site. For the lab's broader coding, reproducibility, and collaboration conventions, see the lab manual itself.

> ⚠ Technology in Rapid Evolution
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> As of early 2026, AI coding assistant technology is changing extremely rapidly, and we are just beginning to figure out how to use these tools effectively ourselves. All information on this site should be taken with extra caution, as best practices and capabilities continue to evolve.

1 Chapters

- **Policies for Using AI**: responsibility for validation, disclosure, attribution, and using AI for journal articles
- **Coding Agents**: what language models and coding agents are; platforms and catalogs; when and how to work with them; prompt formats; and benefits, hazards, and best practices
- **Agent and Harness Architecture**: how harnesses, agent execution loops, and runtime environments are engineered, and the 2026 harness landscape
- **Configuring and Extending Agents**: configuring agent environments, instructions, Agent Skills (`SKILL.md`), MCP servers, plugins, and developer extensions
- **Local Models and Offline Agents**: running agents offline, connecting open harnesses to local Ollama and OpenRouter endpoints, and evaluating small local models
- **Orchestrating Teams of Agents**: when running several agents at once is worth the cost, what we already use for it, outside orchestrators (Agent Teams, Inflexa, TORQCLAW, Oh My OpenCode), and multi-agent frameworks

- **AI for Math and Statistics**: reasoning models for mathematics and statistics, formal proof assistants, and benchmarks
- **Grok Bot and Alternatives**: persistent teammates with their own computer (Grok Bot, Rakazo, OpenClaw), collaborative workspaces (Claude Cowork, Gemini Spark), and hosted agent offices
- **Data Dictionaries for Humans and Agents**: `data-dict.yaml` for making dataset context accessible to both humans and AI agents (see data-dict.tidyverse.org)
- **Benchbook: A Personal Wiki Under Contract**: a plain-markdown, git-versioned wiki that an AI agent maintains under a written contract (see benchbook)
- **Make: Visual Automation and AI Agents**: visual, no-code canvas for 3,000+ apps, scenarios, and AI agents with Make Grid and MCP (see make.com)
- **Pull-Request Workflow with Agents**: filing issues, claiming work, driving a pull request to a clean, mergeable state, and agent-assisted automated code review

The notes are available in multiple formats:

- **HTML Website**: Navigate using the navbar for easy access to all pages
- **RevealJS Slides**: Each chapter can generate a presentation format with `-slides.html` suffix
- **PDF Handouts**: Each chapter can generate a PDF handout with `-handout.pdf` suffix
- **DOCX Documents**: Each chapter can generate a Microsoft Word document with `.docx` extension

2 About this website

This website is built with Quarto, an open-source scientific and technical publishing system, from the UCD-SERG `qwt` (Quarto Website Template).

That template's structure comes from the UCD-SERG Lab Manual, which is in turn adapted from the Benjamin-Chung Lab Manual. This site's own content also began as the "Working with AI" chapter of the UCD-SERG Lab Manual.

3 Building the website

To render the website locally:

```
quarto render
```

To preview the website with live reload:

```
quarto preview
```

The rendered output will be in the `_site/` directory, which is published to GitHub Pages.

4 License

See LICENSE.

References